Do not write outside the box

0 4 . 1 What type of variable is the mass of dissolved solids?

[1 mark]

Tick (✓) **one** box.

Categoric ☐

Control ☐

Dependent ☐

Independent ☐

0 4 . 2 The method produced an error in the mass recorded in step 4.

Suggest what caused the error.

How could the error be avoided?

[2 marks]

Error _____

_____

Avoided by _____

_____

Question 4 continues on the next page

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Another student carried out the investigation correctly.

Table 1 shows the results.

Table 1

Water sample	Mass of dissolved solids in g				
	Test 1	**Test 2**	**Test 3**	**Test 4**	**Mean**
A	0.22	0.23	0.20	**X**	0.21
B	0.03	0.08	0.02	0.03	0.04
C	0.45	0.60	0.49	0.58	0.53
D	0.80	0.91	0.79	0.86	0.84

0 4 . 3 Calculate value **X** in **Table 1**.

[2 marks]

_____

_____

_____

_____

X = _____ g

0 4 . 4 Which water sample has the greatest range of masses of dissolved solids?

Give the reason for your answer.

[2 marks]

Water sample _____

Reason _____

_____

_____

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0 4 . 5 Water companies measure the volume of water used by households in cubic metres (m^3).

25 cm^3 of a different water sample contained 0.016 g of dissolved solids.

Calculate the mass of dissolved solid in 1 m^3 of this water sample.

1 m^3 = 1000 dm^3

Give your answer in standard form.

[4 marks]

Mass (in standard form) = _____ g

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Turn over for the next question